

Kitchcu

Detailed User Flow Encyclopedia - Every Screen, API, Event, Diagram

Kitchcu cloud kitchen platform

Audience: CPO, Product, Engineering, QA, Investors

Three executive lenses:

- Product - goal, persona, entry URL, preconditions, step-by-step UI actions
- Engineering - exact /api/v1 routes, request/response notes, domain events
- QA - success screens, failure paths, order status state machine

This guide includes:

- 9 major flows: owner onboarding, daily ops, customer checkout, multi-kitchen
- split settlement, GST close, admin ops, customer login, coupons, live stream
- Order status state machine + JWT auth types table
- Gateway proxy map + cross-links to API.md, Complete Guide, UI screenshots
- Every route/event traced directly from services/*/app source, July 2026

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- 2. Owner daily ops: login -> WhatsApp/manual order -> lifecycle -> tracking
- 5. Owner GST: profile -> sync -> report -> close audit

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- 3. Customer checkout: discover -> menu -> quote -> pay -> track -> rate
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Purpose, Personas & Auth

This pack is the single source of truth for how a user actually moves through KitchCu: every UI step, every API call, every domain event, every failure path. It expands Complete Guide Part IV (17.1-17.8) into full encyclopedia depth, plus three flows the Guide only summarizes: multi-kitchen split settlement, coupons/CRM, and live streaming opt-in.

Question	Answered by
What is KitchCu / why built this way?	CKAC-COMPLETE-GUIDE.md Parts 0-III
Step-by-step flows, encyclopedia depth?	This document (CKAC-USERFLOWS.md/.pdf)
Exact request/response shape?	API.md; Gateway /docs /redoc /openapi.js
What does the screen look like?	docs/assets/ui/ screenshots
What is built vs. designed?	CKAC-IMPLEMENTATION-GUIDE.md

"Read order: Complete Guide Parts 0-III -> this pack -> API.md for exact schemas -> live /openapi when writing code against a specific route."

2. Persona Surfaces

Persona	Surface	Host : Port	JWT type
Guest / prospect	Portal	kitchcu.in : 13000	none
Owner / chef	Kitchen	kitchen.kitchcu.in : 13002	owner
Customer / diner	Customer	customer.kitchcu.in : 13001	customer
Platform admin	Admin	admin.kitchcu.in : 13003	admin

Edge / service	Port(s)	Role
API Gateway	18000	Sole public HTTP edge for /api/v1/*; pat
identity..streaming (12 services)	18001-18012	Domain services; never called directly b
PostgreSQL 16 + PostGIS	15432	ckac_<domain> schema per service
Redis 7	16379	Streams (events) + tenant-scoped cache

3. Auth JWT Types & Demo Credentials

Type	Issued via	Validated by	Notes
owner	auth/otp/verify	get_current_owner	Dev OTP fixed 123456
customer	customer/whatsapp/verify or oauth/comple	get_current_customer	type==customer checked
admin	admin/auth/login	get_current_admin	Platform scope only
internal	shared secret (not JWT)	X-Internal-Key dependency	Gateway blocks /internal/* (404)

Role	Identifier	Secret	Notes
Owner (primary)	9876543210	OTP 123456	Raj Sharma - CKPNQ001, Pune
Owner	9876543211	OTP 123456	Priya Mehta - Mehta Tiffins
Owner	9876543212	OTP 123456	Amit Desai - Desai Cloud Kitchen
Owner	9876543213	OTP 123456	Sneha Kulkarni - Kulkarni Home Food
Customer	9123456789	OTP 123456	Priya Customer - default diner
Customer	9123456780	OTP 123456	Rahul Menon - repeat/VIP segment
Customer	9988776655	OTP 123456	Ananya Guest - guest checkout
Admin	admin@kitchcu.dev	admin123456	Platform scope only

4. Order Status State Machine

Enforced server-side by can_transition() in services/order/app/models.py. cancelled is reachable from any non-terminal state; delivered/cancelled are terminal.

```
      received -> accepted -> preparing -> ready -> out_for_delivery -> delivered
|           |           |           |           |
+-----+-----+-----+-----+-----+--> cancelled (terminal)
```

Every transition writes `ckac_orders.order_status_events` (immutable audit trail) and publishes `order.status.changed` on `ckac:orders:order`. Notification consumes this for WhatsApp status pushes (F45) and tracking-interval reminders (F29) -- never a fabricated ETA, only owner-set prep/delivery windows.

Onboarding, Daily Ops & GST

Goal: zero to same-day revenue in one sitting. Persona: Owner. Entry: kitchen.kitchcu.in login/register.

```

1. POST /owners/register (new owner)
2. POST /auth/otp/request -> POST /auth/otp/verify -> owner JWT
3. POST /kitchens -> kitchen.created (code CKxxxxnnn)
4. POST /kitchens/{id}/media/upload (live-capture hero, getUserMedia)
5. POST /kitchens/{id}/dishes -> dish.created
  (server REJECTS heroes with is_live_capture:false)
6. POST /kitchens/{id}/orders/manual -> order.placed (status=received)
7. PATCH /orders/{id}/status {accepted} -> order.status.changed
  -> internal deduct-order call to Catalog -> ingredient.stock.deducted
8. GET /kitchens/{id}/analytics/summary -> same-day revenue visible

```

Failure paths

- Wrong/expired OTP -> 401; unregistered phone -> 404 on verify
- Gallery photo as hero -> 400/422, UI blocks save, prompts camera capture
- Invalid status jump (received->ready) -> 400
- Low/zero mapped-ingredient stock on accept -> 200 with stock-warning banner (not blocked)

2. Flow 2 -- Owner Daily Login + Order Intake + Lifecycle + Tracking

Goal: steady-state daily ops loop. Persona: Owner. Entry: kitchen.kitchcu.in dashboard.

```

Customer WA message -> Meta webhook -> Notification
-> POST /internal/kitchens/{id}/orders/from-whatsapp (X-Internal-Key)
-> order_drafts row -> order.draft.created
Owner: GET ../orders/drafts -> review -> POST ../drafts/{id}/confirm
-> order.placed (status=received) [manual alt: POST ../orders/manual]
Owner advances lifecycle: PATCH /orders/{id}/status at each stage
received->accepted->preparing->ready->out_for_delivery->delivered
each hop -> order.status.changed -> Notification: WA push (F45) +
tracking-interval reminders (F29, ckac:notify:tracking)
Public: GET /delivery/track/{token} -> current stage, no fake ETA

```

Failure paths

- Confirm draft referencing a deleted dish -> 400/409, owner edits line items first
- Skipping a lifecycle stage (preparing->delivered) -> 400
- Expired/invalid tracking token -> 404 on public page

3. Flow 5 -- Owner GST: Profile -> Sync -> Report -> Close Audit

Goal: delivered orders -> GST-compliant invoices -> closed monthly audit for accountant handoff. GST registration is optional; dormant until is_active=true.

```

PUT /kitchens/{id}/gst/profile -> gst.profile.created/updated
POST /kitchens/{id}/gst/sync?year&month -> gst.invoice.created x N
GET ../gst/reports/monthly + GET ../gst/reports/balance-sheet
GET ../gst/audit (pre-close review)
POST ../gst/audit/close -> gst.audit.closed (immutable snapshot)

```

Failure paths

- Malformed GSTIN -> 422 validation
- Sync before registration date -> zero new invoices (documented, not an error)
- Closing an already-closed month -> 409; close is one-way (no un-close in v1)

Checkout, Multi-Kitchen & Login

Goal: core trust-to-purchase loop. Persona: Customer. Entry: customer.kitchcu.in.

```

GET /kitchens/public/nearby?diet&live_capture&live_only (PostGIS-sorted)
GET /kitchens/{id}/menu (cached menu:{kitchen_id}, TTL 5 min)
POST /delivery/quote -> fee shown BEFORE payment (closes C2)
  status: ok | out_of_range -- checkout blocked cleanly if out of range
[optional] POST /marketing/coupons/validate
POST /kitchens/{id}/orders/customer -> order.placed + delivery.tracking_created
POST /billing/payments/customer -> .../upi-intent -> .../{id}/capture
  -> payment.created -> payment.captured
Owner advances lifecycle (Flow 2) ... GET /delivery/track/{token}
delivered -> POST /customers/me/orders/{id}/ratings
  (verifies delivered + ownership) -> rating.created / rating.aggregate.updated
overall = 0.6*home_taste + 0.4*quality

```

Failure paths

- Distance beyond max_delivery_radius_km -> out_of_range, checkout blocked, no silent fee inflation
- Payment capture fails/times out -> order stays received/payment pending, retry same payment_id
- Rate a non-delivered or not-owned order -> 403/400

2. Flow 4 -- Multi-Kitchen Cart -> Master Order -> Split Settlement -> Master Receipt

Goal: one cart across 2+ kitchens, one payment, fair independent per-kitchen payout. Kitchen A never sees Kitchen B was in the same cart.

```

Cart grouped by kitchen_id
POST /customers/me/master-orders (Idempotency-Key header)
  -> ATOMIC: any invalid group rolls back the WHOLE master order
  -> master_order.created (MORD-YYYYMMDD-XXXX) + order.placed per sub-order
POST /billing/payments/customer/master -> one aggregated payment
POST .../master/{payment_id}/capture
  -> payment.captured + payment.split.completed
  -> settlement.created per kitchen (net_to_owner, Route-style, NO take-rate)
GET /customers/me/master-orders/{id}/bill.pdf -> ONE master receipt
Each owner: GET /orders/{sub_order_id} -> sees only their own sub-order

```

Failure paths

- One group has an unavailable dish -> 400/409, entire request fails, nothing partially created
- Duplicate submit with same Idempotency-Key -> returns original MasterOrderResponse
- Split capture fails mid-way -> no partial settlement marked paid, retried on same payment_id

3. Flow 7 -- Customer WhatsApp OTP / OAuth Login

Goal: least-friction auth via whichever channel the diner already uses.

Path	Step 1	Step 2	Step 3
WhatsApp OTP	POST .../whatsapp/request	OTP 123456 (dev)	POST .../whatsapp/verify -> customer JWT
Social OAuth	GET .../oauth/providers	GET .../oauth/{p}/start	POST .../oauth/{p}/complete -> customer

customer.created (ckac:identity:customer) fires only on first-time provisioning, from either path. Wrong/expired OTP -> 401; OAuth denial/state mismatch -> login failed, no partial customer record created.

Admin, Coupons & Live Stream

Goal: platform-scope oversight -- is the platform healthy, who is stuck -- never owner-scope menu/order mutation.

```
POST /admin/auth/login          -> admin JWT (type=admin)
GET /admin/stats                -> platform counters
GET /admin/owners | /admin/kitchens | /admin/orders -> platform-wide lists
PATCH /admin/kitchens/{id}/status -> moderate (suspend etc.)
GET /admin/tickets -> GET /admin/tickets/{id} -> PATCH ... -> POST .../reply
-> support.ticket.created / .updated / .replied (ckac:notify:support)
```

Admin JWT used against an owner-mutation route -> 403 by design (scope is intentionally narrow). Wrong password -> 401.

2. Flow 8 -- Coupon Apply / CRM Promotion Path

Goal: owner runs a promotion from their own customer data; customer redeems at checkout -- no cross-tenant data sale, no commission.

```
Owner: GET /kitchens/{id}/crm/customers -> tag segment (PATCH .../crm/customers/{cid})
Owner: POST /kitchens/{id}/coupons      -> coupon.created
Owner: POST /kitchens/{id}/promotions   -> promotion.created
Customer: GET /kitchens/{id}/promotions/active (banner on menu)
Customer: POST /marketing/coupons/validate -> valid + discount_amount, or reason
(checkout continues per Flow 3 with discounted total)
```

Note: no separate promotion-redeem-at-checkout endpoint exists -- promotions surface passively via promotions/active; only coupons have an explicit checkout-time validate call. CRM tag on a customer who never ordered here -> 404.

3. Flow 9 -- Live Stream Opt-In (Go-Live / Live Filter)

Goal: owner opt-in live prep sessions (never mandatory); customers filter discovery to kitchens live right now -- reinforces live-capture trust.

```
Owner: PATCH /kitchens/{id}/stream/settings -> stream.settings_updated
Owner: POST /kitchens/{id}/stream/go-live   -> stream.started
Customer: GET /kitchens/public/nearby?live_only=true
(alt) GET /stream/live-kitchens
Customer: POST /stream/sessions/{id}/viewer-token -> LiveKit viewer join
Owner: POST /kitchens/{id}/stream/end       -> stream.ended
```

- go-live without opt-in enabled -> 403
- go-live while a session is already active -> 409, UI shows existing session
- viewer-token for ended/nonexistent session -> 404

Gateway Map & Cross-Links

Public clients only ever call the gateway (port 18000). `resolve_service_url()` routes by path prefix/marker to the owning service -- never by client-controlled host.

Path prefix / marker	Upstream service
/api/v1/auth/*, /owners/*, /customers/*	identity
/api/v1/customers/me/orders*, master-ord	order
.../customers/me/orders/.../ratings	ratings
/kitchens/* + categories menu dishes cui	catalog
/kitchens/* + orders analytics	order
/kitchens/* + ratings suggestions	ratings
/kitchens/* + crm coupons promotions	marketing
/kitchens/* + gst	billing
/kitchens/* + growth learning commun	growth / learning / community / streamin
/kitchens/* (fallback base CRUD)	identity
/billing/*, /webhooks/razorpay	billing
/marketing/*, /delivery/*, /growth/*, /l	matching service
/webhooks/* (non-razorpay), /support/*,	notification
/internal/*	NOT PROXIED -- gateway returns 404

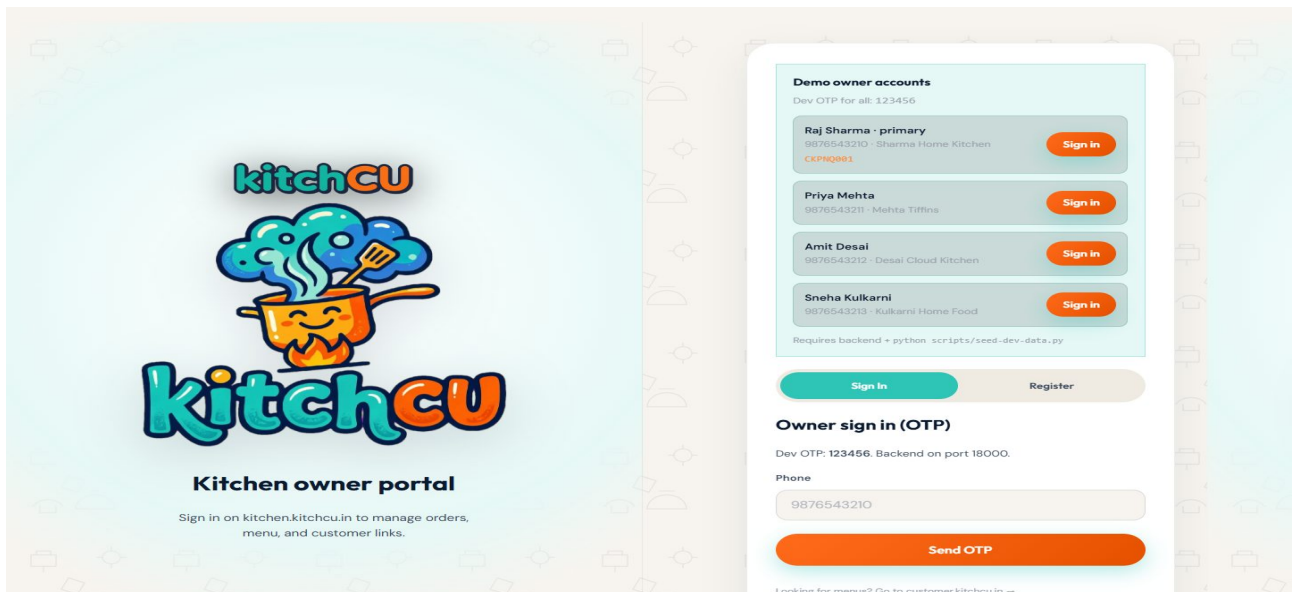
Gateway-owned (not forwarded): GET /, /health/live, /health/ready, /openapi.json (aggregated, ?refresh=true), /docs, /redoc. Portal /openapi renders this same aggregated schema for non-technical browsing.

2. Cross-References

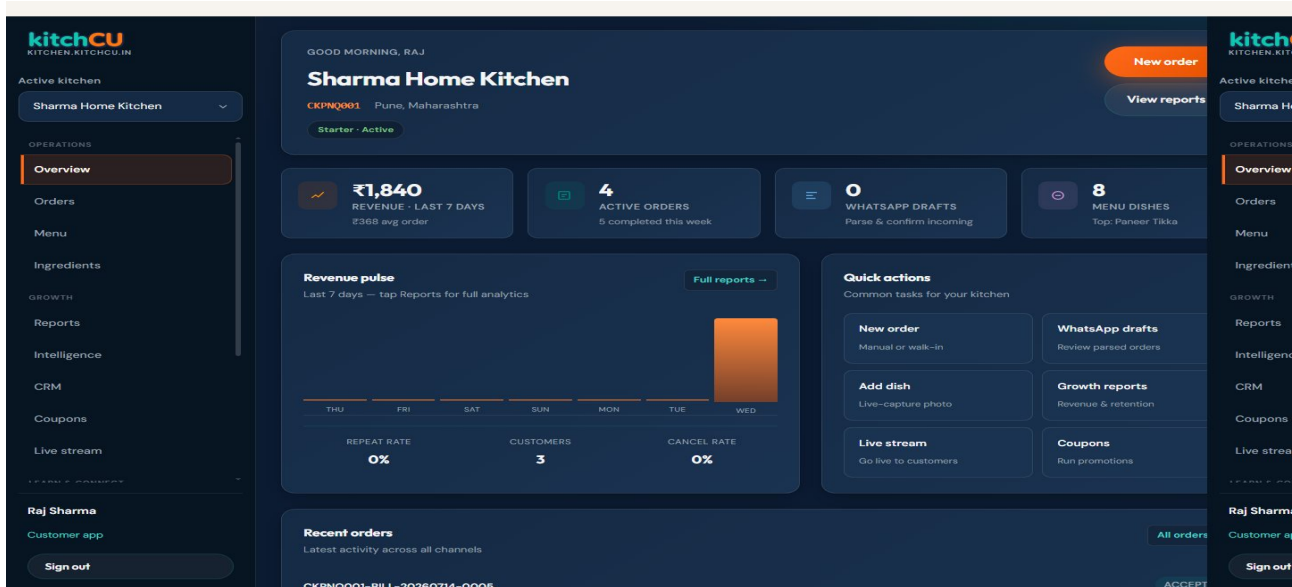
Need	Go to
Exact request/response JSON	docs/API.md; gateway /docs /redoc /opena
CEO/CPO/CTO narrative	CKAC-COMPLETE-GUIDE.md 17.1-17.8 (conden
UI reference screenshots	docs/assets/ui/ (5 surfaces)
Feature acceptance criteria F01-F48	CKAC-COMPLETE-PLANNING-BENCHMARK.md
Architecture / scale / TDD+EDD	CKAC-ARCHITECTURE-CTO.md, KITCHCU-ENGINE
Build status per module	CKAC-IMPLEMENTATION-GUIDE.md

3. UI Reference - Flows in Context

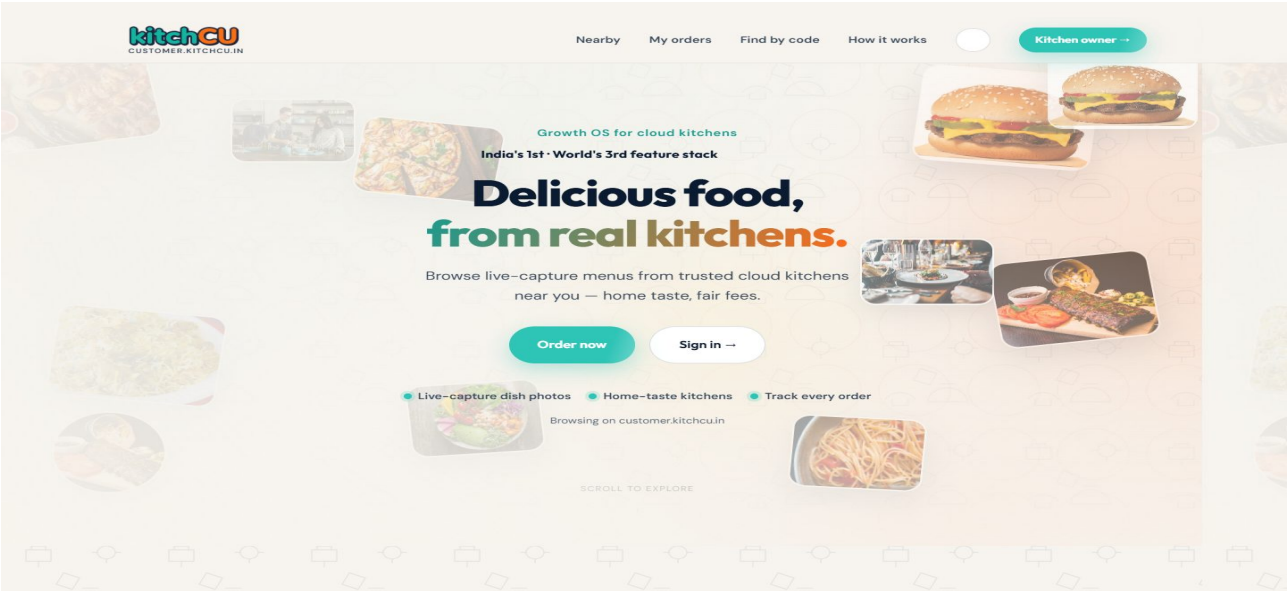
Kitchen login (kitchen.kitchcu.in :13002) -- entry point for **Flow 1** (owner onboarding) and **Flow 2** (daily login). Demo: 9876543210 / OTP 123456.



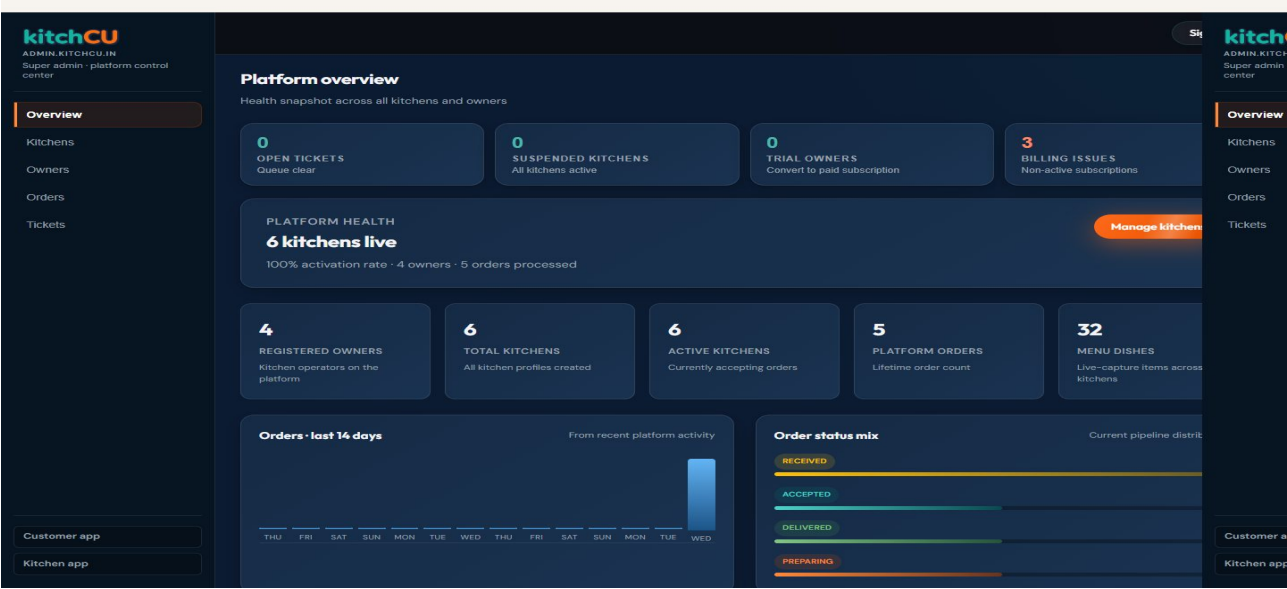
Owner dashboard -- lands here after Flow 1 step 3 and is the home base for Flow 2 (order intake/lifecycle) and Flow 5 (GST) side-nav entries.



Customer home (customer.kitchcu.in :13001) -- entry point for Flow 3 (single-kitchen checkout) and Flow 4 (multi-kitchen cart). Demo: CKPNQ001 near Koregaon Park.



Admin overview (admin.kitchcu.in :13003) -- entry point for Flow 6. Demo: admin@kitchcu.dev / admin123456.



4. Document Control

Field	Value
Document	CKAC-USERFLOWS.md / .pdf
Version	1.0
Date	July 2026
Traceability	Every route/event read directly from ser
Change policy	Update .md whenever a route/event/status
Supersedes	Nothing (net-new pack); condenses into C

"KitchCu User Flow Documentation Pack v1.0 - Confidential - July 2026."